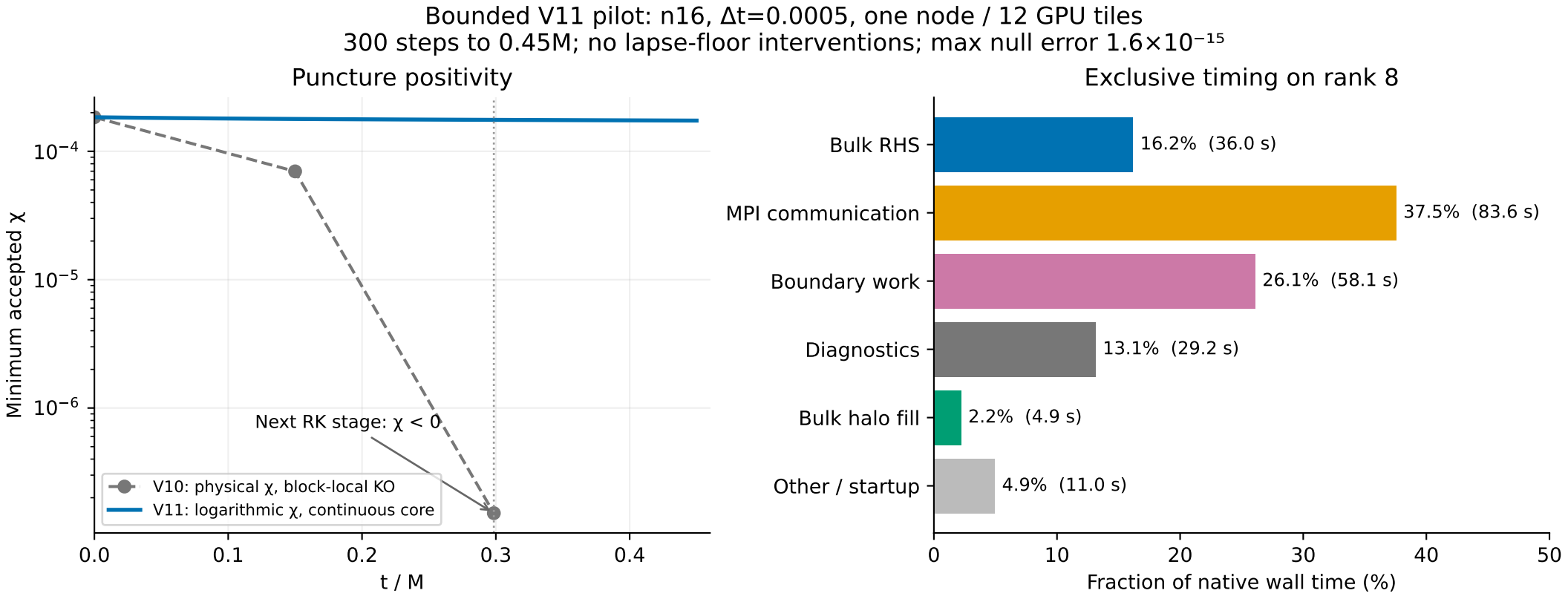
Bounded V11 pilot: n16, Δt=0.0005, one node / 12 GPU tiles
300 steps to 0.45M; no lapse-floor interventions; max null error 1.6×10⁻¹⁵
Puncture positivity
Minimum accepted χ
10⁻⁴
10⁻⁵
10⁻⁶
Next RK stage: χ < 0
V10: physical χ, block-local KO
V11: logarithmic χ, continuous core
0.0
0.1
0.2
0.3
0.4
t / M
Exclusive timing on rank 8
Bulk RHS
16.2% (36.0 s)
MPI communication
37.5% (83.6 s)
Boundary work
26.1% (58.1 s)
Diagnostics
13.1% (29.2 s)
Bulk halo fill
2.2% (4.9 s)
Other / startup
4.9% (11.0 s)
0
10
20
30
40
50
Fraction of native wall time (%)